

Reasoning about Bounded Reasoning

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Motivation

- ▷ Bounded-reasoning models (level-k, CH) are widely used to organize behavior in GT experiments, especially through inferred distributions of “levels”
 - ▷ Strategic reasoning is central, but Nash’s correct-belief premise is hard to defend
 - e.g. initial play, no precedents, no salient benchmark

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 - ▷ a belief-driven bound: beliefs about opponents’ levels
 - ▷ a cognitive bound: limited higher-order reasoning about rationality

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- ▷ But these models conflate two distinct “dimensions”
 - ▷ a belief-driven bound: beliefs about opponents’ levels
 - ▷ a cognitive bound: limited higher-order reasoning about rationality
- ▷ Our question: How can we think about these two dimensions? How do they interact?
 - ▷ what can we identify from observed behavior?

Conceptual Idea

- ▷ We start from a static **complete-information game**
- ▷ We then “lift” it to an **incomplete-information game**
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 - ▷ players are uncertain about opponents’ types
 - ▷ these types play the role of “levels”
- ▷ This separates two objects that exactly capture the two dimensions
 - ▷ **type index** k : beliefs about opponents’ types
 - ▷ **reasoning order** n : higher-order belief in rationality
- ▷ Key idea:
 - ▷ first-order belief restrictions capture bounded reasoning about opponents’ types
 - ▷ bounds on higher-order belief in rationality capture cognitive limits

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- ▷ Three benchmark cases
 - ▷ **Downward rationalizability**: robustness benchmark
 - ▷ **L-rationalizability**: bridge to the classic level- k
 - ▷ **CH-rationalizability**: corresponding bridge for CH
- ▷ Main message
 - ▷ the framework clarifies what level- k -type behavior does and does not identify
 - ▷ clarifies and bridges two approaches to strategic reasoning in games

Two Approaches to Bounded Reasoning

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 - ▷ **Behavioral/procedural**: level- k , Cognitive Hierarchy
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 - ... and provides nuanced, but explicit, reasoning-based foundations of level- k , CH
 - ... and is a conceptually faithful representation of a “level- k world”
 - ... and provides a tractable framework for other, novel, explorations for such a world

Agenda

1. Illustrative Example for Level- k Thinking
2. Model set-up and background knowledge (not today)
3. Level- k Rationalizability (not today)
4. CH Rationalizability (not today)
5. Downward Rationalizability (not today)

1 – Illustrative Example

Illustrative Example

Player 1 \ 2	l	c	r
U	3, 2	2, 1	1, 0
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▷ Sophisticated Strategic Reasoning:

▷ If players are rational

$$R^1 = A_1 \times \{l, c\}$$

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$$R^2 = \{U, M\} \times \{l, c\}$$

- ▷ $R^3 = \{U, M\} \times \{l\}$

- ▷ $R^4 = \{(U, l)\} = R^\infty$

- ▷ Behavioral implications of RCBR

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▷ More generally, $L^k[p] \subseteq R^k$
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▷ level- k is consistent with assumptions
about rationality and higher-order
reasoning about rationality

Our Approach for Level- k Thinking

- ▷ Uncertainty about **actions and types** of co-players. **TWO** dimensions to keep track of:
 1. The **type** (aka level) of a player, denoted by k
 2. **Orders of reasoning about rationality** of a player, denoted by n
 - ▷ $n = 1$: Player is rational
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- ▷ Also impose restrictions on what players can think. Roughly:
 1. If you think you might playing against a level 0, think they play according to the **anchor**
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- ▷ Consistency: Restrictions are **transparent** (aka “common knowledge”) among the players
 - ▷ i.e. an instance of Δ -Rationalizability (Battigalli and Siniscalchi), but can be relaxed

L-Rationalizability: Example

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n/k	R^n	$L^k[p]$
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Main Insight for Level- k Thinking

- ▷ Our result:
 - ▷ If players' reasoning depth is **limited**, predictions reduce to *n-rationalizable actions only*
 - ▷ Only once reasoning is **deep enough**, the model delivers the exact level- k predictions

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 - ▷ Only once reasoning is **deep enough**, the model delivers the exact level- k predictions
- ▷ Implication:
 - ▷ A **foundation** (“folk understanding”) for the level- k model within epistemic game theory
 - ▷ Offers a more **nuanced view**: level- k predictions require sufficient higher-order reasoning
 - ▷ Our framework makes this connection **transparent and relatively simple**, while recent approaches deliberately remain within complete information, which leads to more technically involved arguments (*cf.*, Brandenburger, Friedenberg, Kneeland, *work-in-progress*; BFK)
 - ▷ We hope that the **tractability** speaks both to **formal foundations** in epistemic game theory and to the broader **behavioral and experimental economics** community

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 - ▷ If you are of level k , think you face a co-player of **lower a level**, i.e. of level $< k$
Robust approach similar to robust mechanism design/informational robustness and “broadly downward looking” → ***Downward Rationalizability***
 - ▷ If you are of level k , think you face a co-player of **lower a level** drawn from a common, but truncated, distribution
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Akin CH model → ***CH Rationalizability*** with similar insights as for level- k model
- ▷ **Many other options possible** in a tractable manner, to be explored not only theoretically, but also guided by **behavioral and experimental questions**

That's it!

Thank you!

Comments and questions much appreciated!

Gabriel Ziegler, FU Berlin & Berlin School of Economics, www.gabriel-ziegler.com